Electrical energy = power × time of usage = P × t

$= 75 \text{ watt} \times 240 \text{ hour}$

$= 18000 \text{ watt hour}$

$= 18 \text{ kilowatt hour} = 18 \text{ kWh}$

1 electrical unit = 1 kWh

Electrical energy = 18 unit




Incandescent lamps glow for 1000 hours. CFL lamps glow for 6000 hours. But LED lamps glow for 50000 hrs (almost 25 years at 5.5 hour per day).

### 4.3.3 Relation between power and velocity

The work done by a force $\vec{F}$ for a displacement $d\vec{r}$ is

$$W = \int \vec{F} \cdot d\vec{r} \qquad (4.40)$$

Left hand side of the equation (4.40) can be written as

$$W = \int dW = \int \frac{dW}{dt} dt$$

(multiplied and divided by dt) (4.41)

Since, velocity is $\vec{v} = \frac{d\vec{r}}{dt}$; $d\vec{r} = \vec{v}\, dt$. Right hand side of the equation (4.40) can be written as

$$\int \vec{F} \cdot d\vec{r} = \int \left( \vec{F} \cdot \frac{d\vec{r}}{dt} \right) dt = \int \left( \vec{F} \cdot \vec{v} \right) dt \quad \left[ \vec{v} = \frac{d\vec{r}}{dt} \right] \qquad (4.42)$$

Substituting equation (4.41) and equation (4.42) in equation (4.40), we get

$$\int \frac{dW}{dt} dt = \int \left( \vec{F} \cdot \vec{v} \right) dt$$

$$\int \left( \frac{dW}{dt} - \vec{F} \cdot \vec{v} \right) dt = 0$$

This relation is true for any arbitrary value of dt. This implies that the term within the bracket must be equal to zero, i.e.,

$$\frac{dW}{dt} - \vec{F} \cdot \vec{v} = 0$$

Or

$$\frac{dW}{dt} = \vec{F} \cdot \vec{v} = P \qquad (4.43)$$

## EXAMPLE 4.19

A vehicle of mass 1250 kg is driven with an acceleration 0.2 $\text{m s}^{-2}$ along a straight level road against an external resistive force 500 N. Calculate the power delivered by the vehicle's engine if the velocity of the vehicle is 30 $\text{m } s^{-1}$.

### *Solution*

The vehicle's engine has to do work against resistive force and make vechile to move with an acceleration. Therefore, power delivered by the vehicle engine is

P = (resistive force + mass × acceleration) (velocity)

$P = \vec{F}_{-\text{tot}} \cdot \vec{v} = (F_{\text{resistive}} + F)\vec{v}$

$P = \vec{F}_{\text{tot}} \cdot \vec{v} = (F_{\text{resistive}} + ma)\vec{v}$

$= (500 \text{ N} + (1250 \text{ kg}) \times (0.2\ ms^{-2}))$ $(30 \text{ ms}^{-1}) = 22.5 \text{ kW}$



UNIT-4(XI-Physics_Vol-1).indd 192 05-01-2022 22:20:18